

Benjamin Curis-Friedman

438-346-5109 | bencf.dev | benjamin.curis-friedman@mail.mcgill.ca | linkedin.com/in/benjamincf | github.com/benjamincf0

EDUCATION

McGill University 3.96 GPA

Montreal, QC

Bachelor of Software Engineering – Minor of Mathematics

Aug. 2024 – Dec. 2027

Awards (6000\$): Alma Mater Entrance Scholarship, Class of Engineering '83 Scholarship

EXPERIENCE

Software Developer Intern

Jan. 2026 – Aug. 2026

Rogue Research Inc

Montreal, QC

- Contributed to well-known open source projects (pyvista & vtk), improving rendering efficiency by **50x**.
- Coded **vertex shaders in GLSL** to enable realistic optical distortion of a scene.
- Engineered a 3D robot emulator application using pyvista, enabling accurate surgery planning.
- Refactored a brain surgery robot **networking protocol packet parser**, making it resilient to corrupt data.
- Implemented two robotic brain stimulation papers, leveraging **bayesian optimization using Pytorch**.
- Implemented the **SSIM image comparison algorithm** for CI UITests using the Apple vImage framework.
- Developed unit tests for existing classes and methods to improve code coverage in Objective-C.

Artificial Intelligence Intern

May 2025 – Dec. 2025

iSMART AI Lab

Montreal, QC

- Coded a **data pre-processing pipeline** in Python to clean raw human vital signals.
- Built and programmed a task labeling device using Arduino and C++ to split recording segments efficiently.
- Created a **50+ hour multi-modal dataset** of recordings from dozens of study participants.
- Developed a **video processing (synchronization & compression) library** using FFmpeg and OpenCV.

CERTIFICATIONS

AWS Certified Solutions Architect - Associate (SAA-C03)

Sept. 2026

Machine Learning Specialization - Stanford Online

Jul. 2025

PROJECTS

Omniclave  | *FastMCP, FastAPI, Auth0, React, Bun, Codex*

May 2026

- Built an MCP server for a student portal, empowering agents to help with homework and emails.
- Integrated OAuth2 to let users login from their favourite MCP client (i.e. Codex / Claude code).
- Automated a secondary login flow using a backend playwright instance to log into Omnivox on a user's behalf.

AI Library From Scratch  | *Python, NumPy, TensorFlow, Keras, PyGame*

Jul. 2025

- Created a neural network library complete with mini-batch gradient descent and activation/cost functions.
- Trained a sequential neural network achieving ~96% test accuracy on MNIST dataset.
- Implemented customizable network and layer shapes for enhanced flexibility and scalability.
- Visualized inference with an interactive real-time digit recognition game using PyGame.

Cheese Manager  | *Java, JavaFX*

Sept. 2025

- Developed an application for a comté cheese distribution business following the MVC pattern in Java.
- Created an aesthetically pleasing user interface with JavaFX using reusable components.
- Collaborated with teammates to create a UML class diagram and state diagram using Umple.

WebChat  | *Firebase, Javascript, Vue.js, HTML, CSS*

Sept. 2023

- Developed a full-stack web messaging platform with authentication to message friends.
- Programmed search and adding friends features with Google Cloud Functions.
- Implemented Firestore security rules to ensure secure communications.

SKILLS

Languages: Python, C, Objective-C, Java, Bash, C++, Typescript & JavaScript, HTML, CSS, R

Frameworks: AWS, Google Firebase, React, Node.js, JUnit, Tailwind, Cocoa, JavaFX

Developer Tools: Git, Github Actions, Docker, CDash, VS Code, XCode, NeoVim, Linux, Subversion

Libraries: NumPy, Matplotlib, PyVista, VTK, OpenCV, pandas, Ax Platforms, SciPy, Keras, TensorFlow

Spoken Languages: English, French